7.7 DC MOTOR

7.7.1 Motor Principle

An electric motor is a machine which converts electrical energy into mechanical energy. Its action is based on the principle that when a current-carrying conductor is placed in a magnetic field, the conductor experiences a mechanical force. The direction of this force is given by Fleming's left-hand rule and magnitude is given by:

$$F = BIl \ (N)$$

B = flux density due to the flux produced by the field winding

I = Magnitude of current passing through the conductor

l = Active length of the conductor

By construction, there is no basic difference between DC generator and DC motor. In fact, the same DC machine can be used interchangeably as a generator or as a motor. In generator the input is mechanical energy by a prime mover and develops a voltage, while in a DC motor the input is electrical energy and it develops rotating torque.

7.7.2 Fleming's Left Hand Rule

Keep the forefinger, middle finger and thumb of the left hand mutually perpendicular to one another. If the forefinger indicates the direction of the magnetic field (B), the middle finger indicates the direction of current (I) in the conductor, and the thumb points to the direction of motion (F) of the conductor.

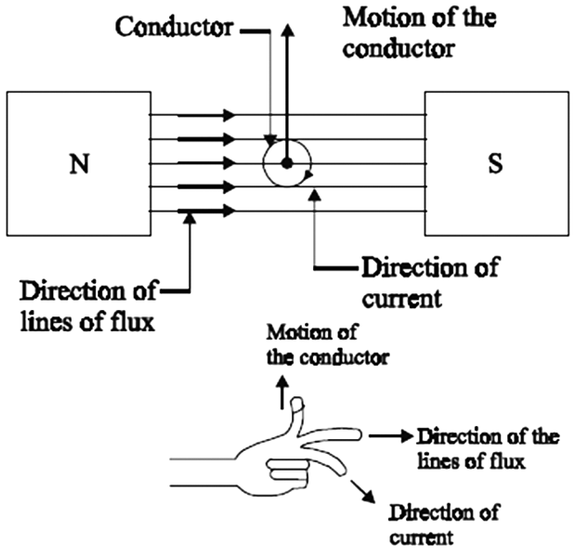


Fig 7.10 *Fleming's Left Hand Rule*

7.7.3 Principle of Operation of DC Motor

A motor is a mechanism by which electrical energy is converted into mechanical energy. Both in principle and design, a DC motor is the reverse process of DC generator.

Consider a single conductor placed in a magnetic field as shown in fig 7.10. The two diagrams indicate the direction of magnetic fluxes considering separately.

How the resultant rotation takes place in a DC motor is clearly understood from the following descriptions in the Fig 7.11.

A steady current is passed through the armature coil from the commutator and the brushes are so arranged as to reverse the current every half revolution. When a coil, carrying a current is placed in magnetic field, it experiences forces, given by Fleming's left-hand rule, which

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